

VMC-3000 Series | MID 130 | Subsystem Code: AXS

Document Number	DOC-EIQ-003	Classification	CONTROLLED
Revision	B	Issued By	EquipmentIQ Technical Publications
Issue Date	06 May 2026	Applies To	EquipmentIQ VMC-3000 Series

1. Introduction

This manual covers the three-axis servo drive and motion control system of the VMC-3000 machining centre (subsystem AXS, MID 130). Each linear axis (X, Y, Z) is driven by a dedicated AC servo motor with absolute encoder, ball screw, and profiled linear guide rail. The servo drives receive interpolated position commands from the CNC-500 controller via FSSB and close the position loop at 1 ms cycle time.

2. Axis Servo Motors

Parameter	X-Axis	Y-Axis	Z-Axis
Model	SVM-1500-X	SVM-1500-Y	SVM-2200-Z
Rated Power	1.5 kW	1.5 kW	2.2 kW
Rated Torque	7.2 Nm	7.2 Nm	10.5 Nm
Peak Torque	21.6 Nm	21.6 Nm	31.5 Nm
Rated Current	6.5 A	6.5 A	9.2 A
Maximum Speed	3,000 RPM	3,000 RPM	3,000 RPM
Encoder	17-bit absolute	17-bit absolute	17-bit absolute
Brake	None	None	24 V DC spring-set
Frame	IEC 90L	IEC 90L	IEC 100L

Table 2-1: Axis Servo Motor Specifications

NOTE: The Z-axis motor includes a spring-set electromagnetic brake (PN: EIQ-AXS-BRK-Z) which engages automatically when the drive is de-energised to prevent axis drop. Do not disable the brake without first supporting the spindle head with a block.

3. Ball Screw & Linear Guide Specifications

Parameter	X-Axis	Y-Axis	Z-Axis
Ball Screw Diameter	32 mm	32 mm	40 mm
Ball Screw Lead	16 mm/rev	16 mm/rev	16 mm/rev
Ball Screw Grade	JIS C3	JIS C3	JIS C3
Ball Screw Nut Type	Double-nut, preloaded	Double-nut, preloaded	Double-nut, preloaded
Max Backlash (spec)	< 0.005 mm	< 0.005 mm	< 0.005 mm
Linear Guide Type	THK SRG 35C	THK SRG 35C	THK SRS 45C
Number of Rails	2	2	2
Guide Preload Class	Z1 (light)	Z1 (light)	Z2 (medium)
Lube Type	ISO VG 32 Way Lube	ISO VG 32 Way Lube	ISO VG 32 Way Lube
Lube Interval	15 min / 2 s pulse	15 min / 2 s pulse	15 min / 2 s pulse

Table 3-1: Ball Screw and Linear Guide Specifications

4. Monitored Parameters

Param ID	PID	Parameter Name	Unit	Normal Range	Critical Range
P010	10	X-Axis Position Error	mm	-0.01 – 0.01	-0.5 – 0.5
P011	11	Y-Axis Position Error	mm	-0.01 – 0.01	-0.5 – 0.5
P012	12	Z-Axis Position Error	mm	-0.01 – 0.01	-0.5 – 0.5
P013	13	X-Axis Servo Current	A	0 – 18	0 – 25
P014	14	Y-Axis Servo Current	A	0 – 18	0 – 25
P015	15	Z-Axis Servo Current	A	0 – 22	0 – 30
P016	16	Axis Following Error	mm	0 – 0.05	0 – 2.0
P017	17	Servo Drive Temperature	°C	20 – 70	0 – 90

Table 4-1: AXS Monitored Parameters

5. Axis Wiring Reference

Table 5-1: Axis Servo System Wiring Connections

Signal Name	From (Module:Pin)	To (Module:Pin)	Wire Gauge	Color Code	Notes
X-Axis Motor U	AXS-DRV-X:U	MTR-X:U1	2.5 mm2	Brown	Screened, 5 m
X-Axis Motor V	AXS-DRV-X:V	MTR-X:V1	2.5 mm2	Black	
X-Axis Motor W	AXS-DRV-X:W	MTR-X:W1	2.5 mm2	Grey	
X-Axis Encoder	MTR-X:ENC	CNC-500:AX1-ENC	0.5 mm2	Violet	17-bit absolute, Cat5e
Y-Axis Motor U	AXS-DRV-Y:U	MTR-Y:U1	2.5 mm2	Brown	Screened, 5 m
Y-Axis Motor V	AXS-DRV-Y:V	MTR-Y:V1	2.5 mm2	Black	
Y-Axis Encoder	MTR-Y:ENC	CNC-500:AX2-ENC	0.5 mm2	Violet	17-bit absolute, Cat5e
Z-Axis Motor U	AXS-DRV-Z:U	MTR-Z:U1	4.0 mm2	Brown	Heavier gauge, 2.2 kW
Z-Axis Motor W	AXS-DRV-Z:W	MTR-Z:W1	4.0 mm2	Grey	
Z-Axis Encoder	MTR-Z:ENC	CNC-500:AX3-ENC	0.5 mm2	Violet	17-bit absolute
Z-Axis Brake	PLC-Q:Y0.5	MTR-Z:BRK+	0.75 mm2	Red	24 V DC, spring-set
X Home Switch	SW-HM-X:NO	PLC-I:X1.0	0.5 mm2	White	NPN proximity, 24V
Y Home Switch	SW-HM-Y:NO	PLC-I:X1.1	0.5 mm2	White	NPN proximity
Z Home Switch	SW-HM-Z:NO	PLC-I:X1.2	0.5 mm2	White	NPN proximity
X Over-travel +	SW-OT-X+:NC	EMG-CHAIN:IN4	0.75 mm2	Orange	Hardwired to E-stop chain
Y Over-travel +	SW-OT-Y+:NC	EMG-CHAIN:IN5	0.75 mm2	Orange	
Z Over-travel -	SW-OT-Z-:NC	EMG-CHAIN:IN6	0.75 mm2	Orange	

6. Error Codes — AXS Subsystem

Error Code	Severity	Title	Primary Param	Action
AXS-CR-001	CRITICAL	Axis Following Error — Emergency Stop	—	Immediate machine stop. Do not restart with
AXS-CR-002	CRITICAL	Servo Drive Hardware Fault — Axis Disabled	—	Immediate machine stop. Do not restart with
AXS-MJ-001	MAJOR	Axis Following Error — Major Limit	—	Stop current operation. Machine must be in
AXS-MJ-002	MAJOR	Servo Drive Thermal Warning — Overtemperature	—	Stop current operation. Machine must be in
AXS-SR-001	SERIOUS	Axis Vibration — Resonance Detected	—	Complete current cycle then stop. Schedul
AXS-SR-002	SERIOUS	Servo Current — Elevated Load on Axis	—	Complete current cycle then stop. Schedul
AXS-MD-001	MODERATE	Following Error — Elevated But Within Limit	—	Log fault. Schedule maintenance within 24
AXS-MD-002	MODERATE	Servo Drive Temperature — Elevated	—	Log fault. Schedule maintenance within 24
AXS-MN-001	MINOR	Servo Current — Minor Variance	—	Log fault. Monitor parameters. Schedule in
AXS-MN-002	MINOR	Axis Vibration — Minor Threshold	—	Log fault. Monitor parameters. Schedule in
AXS-WN-001	WARNING	Following Error — Intermittent Spikes	—	Log event. No immediate action required. R
AXS-NC-001	NOTICE	Axis Position — Reference Established	—	Informational. Parameter approaching limit
AXS-AD-001	ADVISORY	Rapid Override Active	—	Performance advisory only. No maintenanc

Table 6-1: Axis Servo Error Codes

7. Servo Tuning & Commissioning

After any mechanical repair (ball screw replacement, bearing change, guide rail replacement), the affected axis servo loop must be re-tuned using the EIQ servo tuning utility (menu: SYS > SERVO > AUTO-TUNE). The following parameters must be recorded before and after tuning:

- Position loop gain (Kp): nominal 80–120 (machine-dependent).
- Velocity loop gain (Kv): nominal 1500–2500.
- Following error at 1,000 mm/min rapid: must be <0.01 mm steady-state.
- Quadrant spike at reversal: must be <0.005 mm after backlash compensation.
- Servo vibration check: run at F=1000, F=5000 mm/min and verify P016 <0.02 mm.
- Thermal drift over 4-hour warm-up: record axis position shift at G54 datum.

8. Spare Parts — Axis Servo System

Part Number	Description	Qty	Unit	Lead Time	Safety Stock
EIQ-AXS-BSC-X32	Ball screw X-axis 32mm dia, C3, 800mm length	1	Each	6 weeks	1
EIQ-AXS-BSC-Z40	Ball screw Z-axis 40mm dia, C3, 600mm length	1	Each	6 weeks	1
EIQ-AXS-NUT-32	Ball screw nut 32mm double-preloaded (X/Y)	1	Each	4 weeks	2
EIQ-AXS-NUT-40	Ball screw nut 40mm double-preloaded (Z)	1	Each	4 weeks	1
EIQ-AXS-ENC-17B	17-bit absolute encoder (any axis)	1	Each	3 weeks	2

Part Number	Description	Qty	Unit	Lead Time	Safety Stock
EIQ-AXS-BRK-Z	Z-axis electromagnetic brake, 24VDC spring-set	1	Each	2 weeks	1
EIQ-AXS-DRV-15	Servo drive 1.5 kW (X/Y axis)	1	Each	4 weeks	1
EIQ-AXS-DRV-22	Servo drive 2.2 kW (Z axis)	1	Each	4 weeks	1
EIQ-AXS-PROX-24V	NPN proximity switch 24VDC (home/overtravel)	1	Each	1 week	5

Table 8-1: Axis Servo Spare Parts List